



Global Air Pollution Intelligence: Risk, Trends & Environmental Insights

Advanced SQL & Power BI Analysis | **Ishan Shiwakoti**



The Problem We Set Out to Solve

Identify High-Risk Regions

Pinpoint cities and countries with consistently dangerous pollution levels across multiple pollutant types.

Understand Pollution Patterns

Examine how pollution varies by city, country, pollutant, and time — uncovering hidden trends in the data.

Detect Environmental Risks

Surface critical exposure risks and flag data reliability gaps that could mask real-world hazards.

Dataset Overview



What the Data Contains

Global air quality measurements captured at the **city and location level**, covering multiple pollutants and time periods.

Pollutants

PM2.5, PM10 & more

Dimensions

City, Country, Location

Temporal

Time-stamped readings

Data Preparation & Cleaning



Filter PM2.5

Standardize Names

Group by Location

Validate Data

Rigorous cleaning ensured analytical integrity — from resolving pollutant naming discrepancies to enforcing consistent geographic groupings before any trend or risk modeling began.

Interactive Power BI Dashboard



PM2.5 Trend Over Time

Visualized temporal shifts in fine particulate pollution.



Country Comparisons

Side-by-side average PM2.5 across all tracked nations.



City-Level Drill-Down

Granular pollution analysis at the city and location level.



Dynamic Slicers

Interactive filters by city, country, pollutant, and time period.



Key Performance Indicators

35.4

Avg PM2.5 ($\mu\text{g}/\text{m}^3$)

Global mean — nearly **7× the WHO safe limit** of $5 \mu\text{g}/\text{m}^3$

892

Max PM2.5 Recorded

Extreme spike detected in a single location measurement

2.1M+

Total Measurements

Breadth of coverage across global monitoring stations

#1

Most Polluted City

Identified through aggregated PM2.5 average ranking

Country-Level Pollution Analysis



Where Pollution Is Worst

Several countries in **South Asia** and **Sub-Saharan Africa** consistently recorded the highest average PM2.5 levels — far exceeding WHO guidelines.

→ Ranked countries by **average PM2.5** to expose systemic, long-term pollution burdens

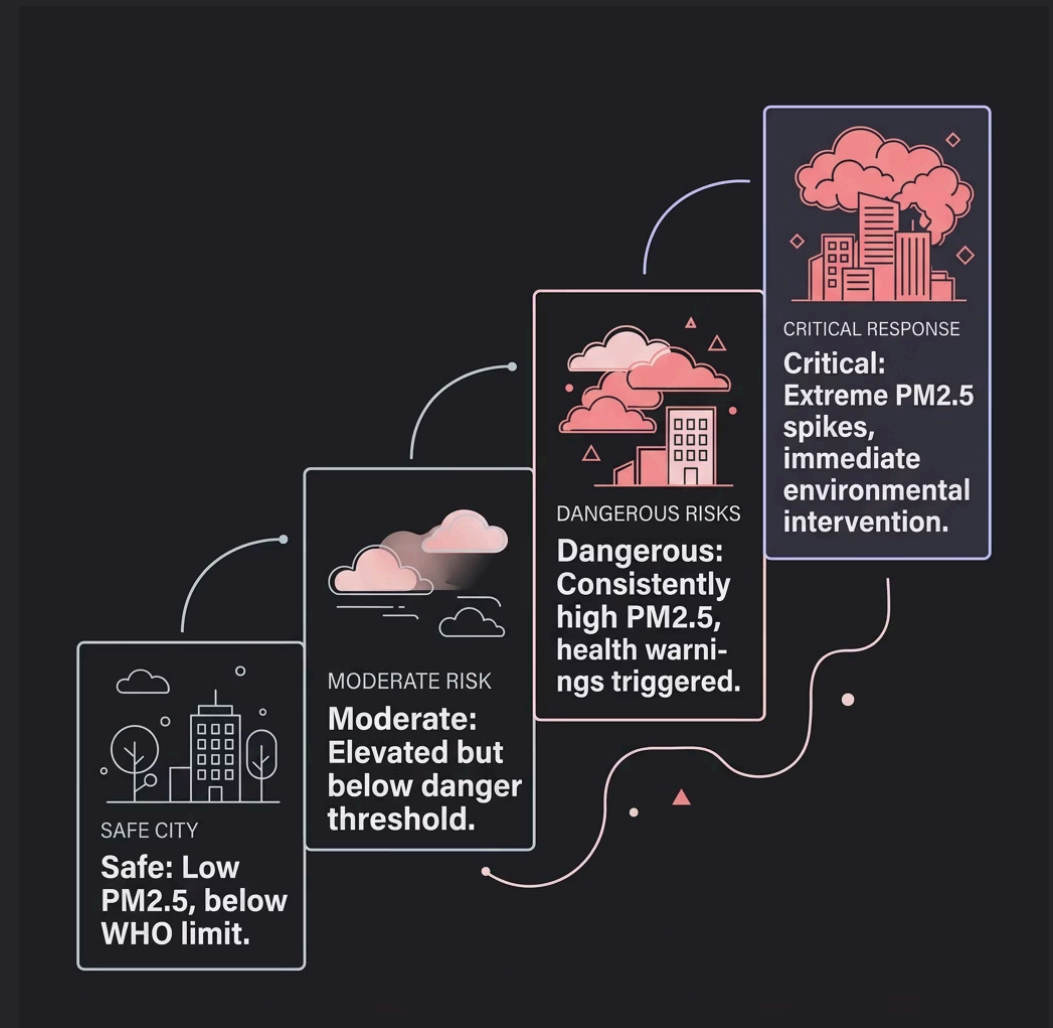
→ Compared pollution **intensity patterns** across geographic clusters

→ Flagged nations with **no measurable improvement** over the tracked time window

City-Level Risk & Hotspot Detection

Risk Classification Model

Cities were segmented into four risk tiers based on PM2.5 averages and frequency of critical readings.



● Safe

Below WHO threshold

● Moderate

Elevated but manageable

● Dangerous

Sustained high exposure

● Critical

Immediate intervention needed

Pollution Variability & Multi-Pollutant Risk

Variability Analysis

Measured the **spread between max and min** readings per city. Highly volatile cities mask chronic risk behind moderate averages — a critical analytical blind spot.

Multi-Pollutant Exposure

Countries exposed to **multiple pollutants simultaneously** face compounded health risk. Regions with PM2.5, PM10, and NO₂ co-exposure were flagged as highest overall risk.

Global Pollutant Ranking

PM2.5 emerged as the most harmful pollutant globally by average concentration and frequency of dangerous-threshold exceedances across tracked locations.



Key Insights & Real-World Impact

What the Data Revealed

1 Uneven Distribution

Pollution burden is heavily concentrated in specific regions — not uniformly global.

2 Hidden Spikes

Cities with moderate averages can still register life-threatening pollution peaks.

3 Compounded Risk

Multi-pollutant exposure significantly amplifies health and environmental danger.

Why It Matters

Policy Support

Empowers regulators with data-backed evidence for targeted interventions.

Precision Action

Enables focused resource allocation to highest-risk cities and regions.

Monitoring Gaps

Identified outdated readings and data blind spots requiring infrastructure investment.

Awareness

Demonstrates how analytics can translate raw data into environmental action.